Infosys Springboard Virtual Internship 6.0 Completion Report

Batch Number - 2

Start date – 04-SEP-2025

Names:

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Internship Duration: 8 Weeks

1. Project Title

ElectViz Election Data Visualization for Media

2. Project Objective

The primary objective of the ElectViz project is to analyze and visualize complex data from Indian state elections using Power BI. The goal is to transform vast and raw election data into a clear, interactive, and easily understandable dashboard.

This tool is designed specifically for media organizations, political analysts, and the public to explore voting patterns, party performance, and historical trends without needing to sift through complex spreadsheets.

The main goals of this project are:

1. To consolidate and analyze key election metrics, including total votes, party participation, seats won, and voter turnout.
2. To identify historical voting trends by visualizing data across different election years.
3. To provide a comparative analysis of political party performance at both the state and national levels.
4. To create a detailed breakdown of seat distribution by state and constituency type.
5. To utilize Power BI's interactive features (slicers, filters, and drill-downs) to empower users to find specific insights dynamically.
6. To provide a data-driven tool for media to support accurate reporting and insightful political commentary.

3. Project description in detail

The ElectViz project was developed as a part of Infosys Springboard Virtual Internship 6.0 to explore how data visualization can enhance political and electoral reporting. Our team used the Election dataset from Kaggle to create a Power BI dashboard that highlights national and regional election results, party performance comparisons, alliance analytics, and voter behavior patterns.

The dashboard includes five main pages: National Overview, State-wise Results, Party Performance, Voter Turnout, and Gender Representation. Each page uses interactive filters, KPIs, and charts to summarize large-scale election results for easy interpretation by journalists and citizens.

Overview and Purpose

In the fast-paced media landscape, especially during election cycles, there is a critical need for quick, accurate, and compelling data insights. Indian elections are famously large and complex, involving thousands of parties and billions of votes across decades.

The ElectViz project addresses this need by creating a single, unified Power BI dashboard. It provides a comprehensive overview of Indian state election history, filtered by year, state, party, and constituency. The dashboard allows media users to instantly identify key statistics, spot trends, and compare party performance, enabling them to craft data-driven stories and reports.

Dataset Description

The dataset used for this project is a comprehensive, granular collection of Indian state election results. Based on the raw data screenshot provided, the dataset includes the following key fields:

- **st_name:** The state where the election was held (e.g., Maharashtra).
- **year:** The year of the election (e.g., 1985).
- **ac_no:** The unique assembly constituency number.
- **ac_type:** The type of constituency (e.g., GEN for General).
- **cand_name:** The name of the candidate.
- **cand_sex:** The gender of the candidate (e.g., M).
- **partyname:** The full name of the political party (e.g., Independent).
- **partyabbre:** The abbreviation for the political party (e.g., IND).
- **totvotpoll:** The total votes polled by the specific candidate.
- **electors:** The total number of eligible voters in the constituency.
- **result:** The outcome for the candidate (e.g., won, lost).
- **Seats Won:** A numeric column indicating if a seat was won (1 for won).

Tools and Technologies Used

1. Microsoft Power BI Desktop: The primary tool used for data modeling, analysis, and creating all dashboard visualizations.
2. Power Query Editor (in Power BI): Used for data cleaning, transformation, and preparing the raw data for analysis (e.g., managing 1,637 party names, standardizing formats).
3. DAX (Data Analysis Expressions): Used to create key measures and KPIs, such as Total Votes, Total Parties, Total Seats, and Voter Turnout %.
4. Microsoft Excel: Likely used as the initial data source for collecting and formatting the raw election data before importing it into Power BI.
5. Canva & PowerPoint – for presentation design and visual enhancement.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Orientation on project goals, understanding Power BI basics, and initial dataset exploration. Team role assignment.	Successfully set up the project workspace, defined the scope of the ElectViz dashboard, and analyzed the structure of the raw election data.
Week 2	Data collection and preprocessing. Cleaning and formatting the large dataset (party names, states, years) using Power Query.	Handled duplicates, standardized over 1,600 party names, and formatted data types. Imported the cleaned dataset into Power BI.
Week 3	Data modeling. Establishing relationships between key tables (States, Parties, Election Results).	Created a robust data model connecting all data points. This was crucial for enabling cross-filtering between slicers and visuals.
Week 4	Develop core DAX measures (Total Votes, Total Parties, Total Seats, Voter Turnout %). Build the main "Indian State Election Analysis" dashboard (Dashboard 1).	Created all primary KPIs using DAX. Designed and completed the main dashboard layout with slicers for Year, Party, State, and Constituency Type.
Week 5	Develop key visuals for historical trends (Dashboard 2), top performers (Dashboard 3),	Added line charts, bar charts, pie charts, and funnel charts to represent historical vote counts,

	and vote share analysis (Dashboard 4).	cumulative seat growth, and party vote shares.
Week 6	Build the detailed "State-wise Party Seat Distribution" matrix (Dashboard 5). Implement and test all interactivity and filters across all pages.	Completed the complex matrix visual. Ensured all slicers and filters correctly updated all 5 dashboard pages simultaneously.
Week 7	Testing and debugging of all visuals, data relationships, and dashboard responsiveness. Mentor review and feedback integration.	Conducted performance checks and data validation. Refined visuals for clarity, improved tooltips, and aligned all dashboard elements for a professional theme.
Week 8	Final presentation and documentation preparation.	Delivered the final interactive ElectViz Power BI dashboard presentation. Compiled this internship completion report and submitted all deliverables.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	The project began with a detailed analysis of the project brief. The team defined the problem statement (visualizing complex election data for media) and finalized the scope for the 5-page dashboard.	15th September , 2025
Prototype/First Draft	The raw election dataset (covering 1977-2014) was successfully cleaned, transformed, and modeled in Power BI. All relationships were established. The first version of the main dashboard (Dashboard 1)	22nd September ,2025

	was completed. This included working KPIs for Total Votes, Parties, and Seats, along with functional slicers.	
Mid-Term Review	All five dashboards were functionally complete. The project was presented to the mentor for feedback, which was then integrated to improve usability and visual clarity.	6th October , 2025
Final Submission	The final, polished ElectViz.pbix file, incorporating all feedback and passing all data validation checks, was completed and submitted.	20th October ,2025
Presentation	The completed project was presented to mentors and peers, demonstrating the dashboard's interactive features and highlighting key insights from the election data.	29th October, 2025

5b. Project execution details

The ElectViz project was executed over an eight-week period, following a systematic process from data preparation to dashboard design, testing, and final presentation. The project's core aim was to transform a highly complex and large dataset of Indian State Election results into an intuitive, interactive tool for media professionals and analysts.

1. Requirement Understanding and Planning

The project began with a brainstorming session to understand the needs of the target audience (media and political analysts). The team identified the key questions they would ask: "Who won?", "Where?", "By how much?", and "What are the trends?". This led to the planning of a multi-page dashboard, with each page answering a specific set of questions. Roles were distributed for data cleaning, DAX measure creation, visualization, and documentation.

2. Data Collection and Preparation

This was one of the most critical phases. The team worked with a granular dataset spanning several decades (1977-2014).

- **Data Cleaning:** The dataset contained over 1,637 unique entries for political parties, including full names, abbreviations, and duplicates. Using Power Query Editor, the team grouped and standardized these party names to ensure accuracy.
- **Transformation:** Data types were corrected (e.g., converting text to numbers). Columns were renamed for clarity (e.g., st_name to State Name).
- **Modeling:** The cleaned data was loaded into Power BI, and a data model was built to link election results with state information, party details, and dates.

3. Dashboard Development in Power BI

The team designed and built a 5-page interactive report, with a consistent dark theme and clear navigation.

- **a) Dashboard 1: Indian State Election Analysis (Main Dashboard)** This page serves as the main summary. It features four key performance indicators (KPIs) at the top: **Total Votes (3044M)**, **Total Parties (1637)**, **Total Seats (5670)**, and **Voter Turnout % (5.52%)**. The main visuals are a horizontal bar chart for "Seats Won by party Name" and a vertical bar chart for "Sum of Seats Won by State," immediately showing the dominance of Independent candidates and states like Uttar Pradesh.
- **b) Dashboard 2: Temporal & Turnout Analysis** This page focuses on historical trends. It features a line chart ("Total Votes by year") that visualizes the fluctuations in voting over decades. A donut chart ("Voter Turnout % by State Name") provides a clear breakdown of voter engagement by state.
- **c) Dashboard 3: Top Performers & Cumulative Growth** This dashboard provides quick insights into the top performers with two large cards: **"Top Party: Independent"** and **"Top State: Uttar Pradesh"**. A waterfall chart ("Count of Seats Won by year") effectively illustrates the cumulative growth of total seats over time.

- **d) Dashboard 4: Share & Distribution Analysis** This page focuses on proportional analysis. A funnel chart ("Count of Seats Won by State name") visually ranks states by seat count. A pie chart ("Vote share by party") shows the percentage of votes for each party, again highlighting the large share of Independent candidates.
- **e) Dashboard 5: State-wise Party Seat Distribution (Matrix)** This is the most granular page, designed for detailed analysis. It consists of a single large matrix visual. It uses **State Name** as rows and **Party Abbreviations** as columns, with the sum of seats won as the values. This allows media users to instantly look up the specific performance of any party in any state.

4. Interactivity and Analytics

The true power of the dashboard comes from its interactivity.

- **DAX Measures:** All KPIs (Total Votes, Total Parties, etc.) were written using DAX to ensure they respond dynamically to user selections.
- **Slicers:** The main dashboard includes slicers for **Election Year**, **Political Party**, **Select State**, and **Constituency Type**. These slicers are synced across all 5 pages. A user can select "1985" and "Bharatiya Janata Party," and all 5 dashboards will instantly update to show data only for that specific context.

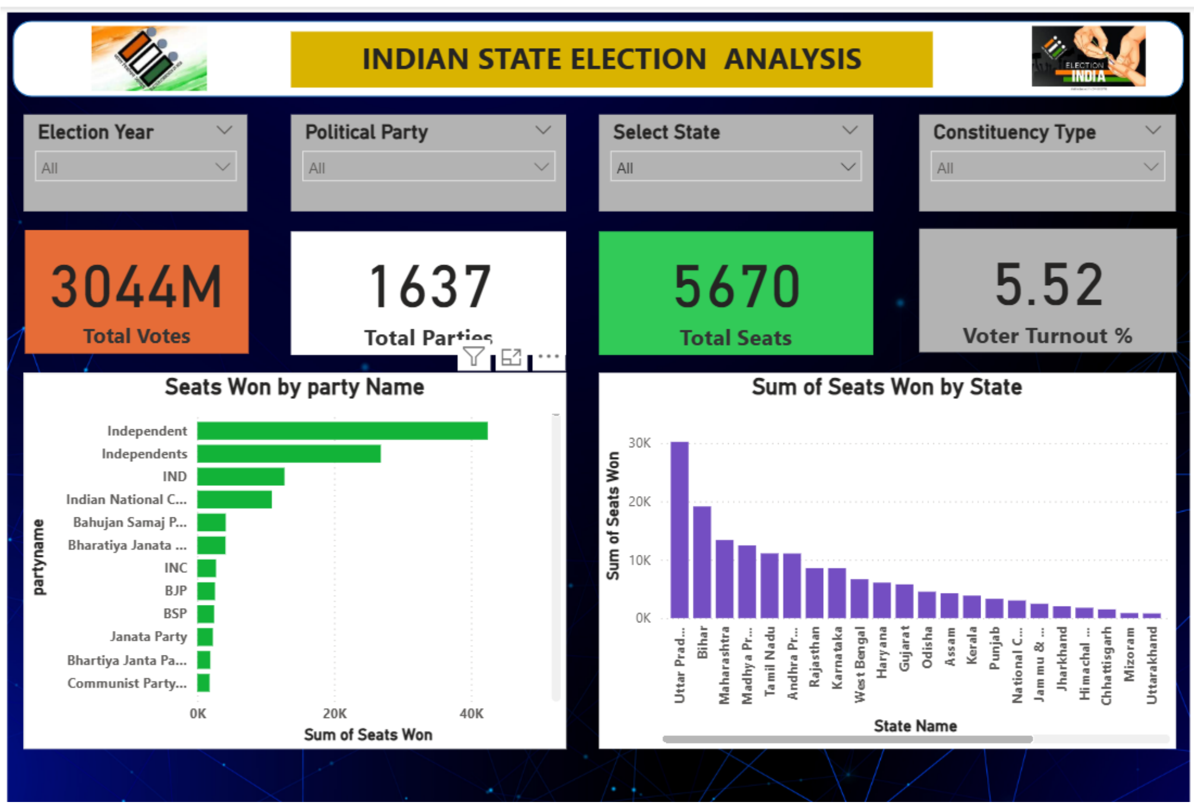
5. Testing and Validation Before finalizing, the dashboard was rigorously tested.

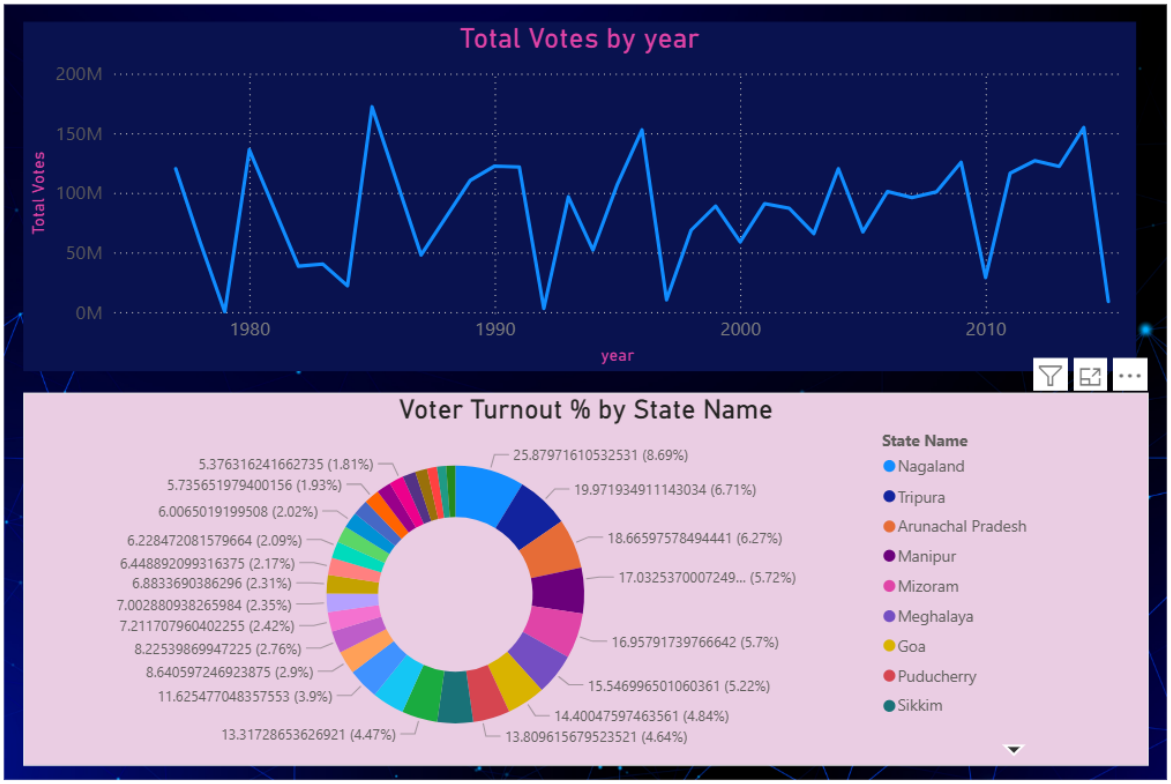
- **Data Validation:** The team cross-checked the KPI values against the raw data to ensure accuracy. For example, the sum of all seats in the Dashboard 5 matrix was validated against the "Total Seats" KPI.
- **Functional Testing:** All slicers, filters, and chart interactions were tested to ensure they worked as expected and did not "break" any visuals.
- **Visual Optimization:** Charts were aligned, and text sizes were standardized for a clean, professional look.

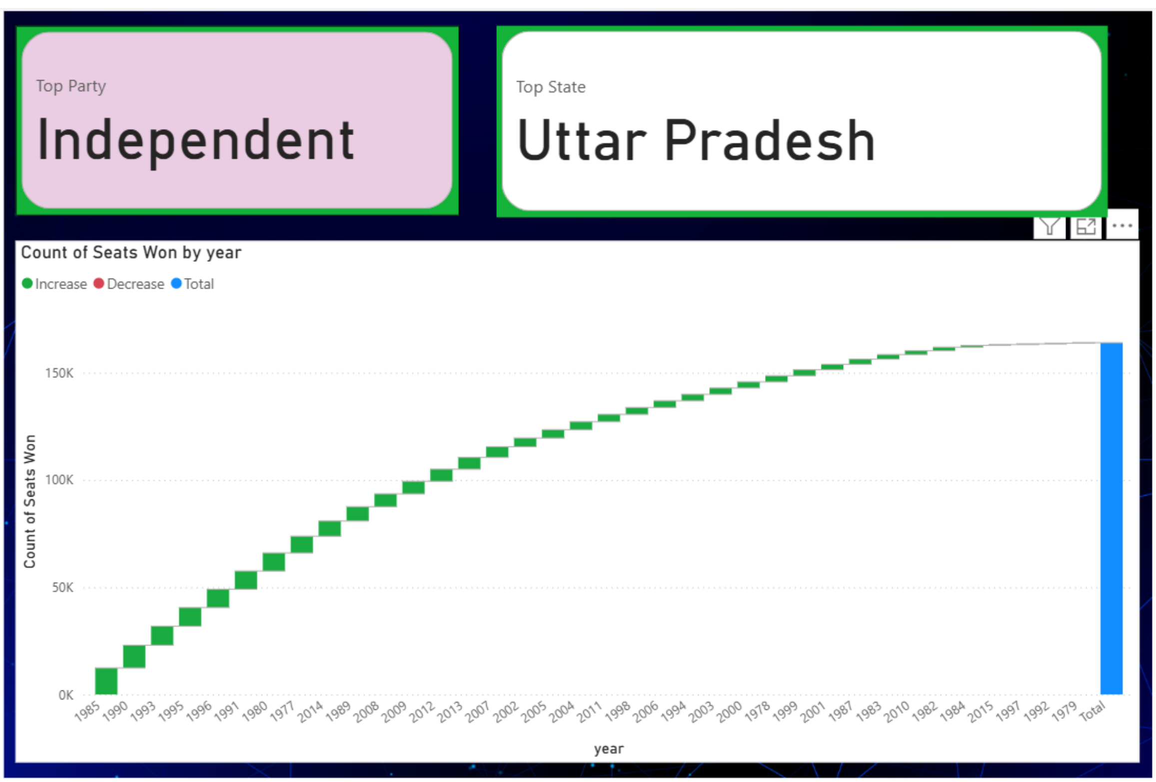
6. Review, Refinement, and Final Presentation All five dashboards were functionally complete and presented to the mentor. Based on the feedback, the team integrated changes to improve usability and visual clarity, such as enhancing tooltips on charts to provide more context and ensuring the color theme was consistent.

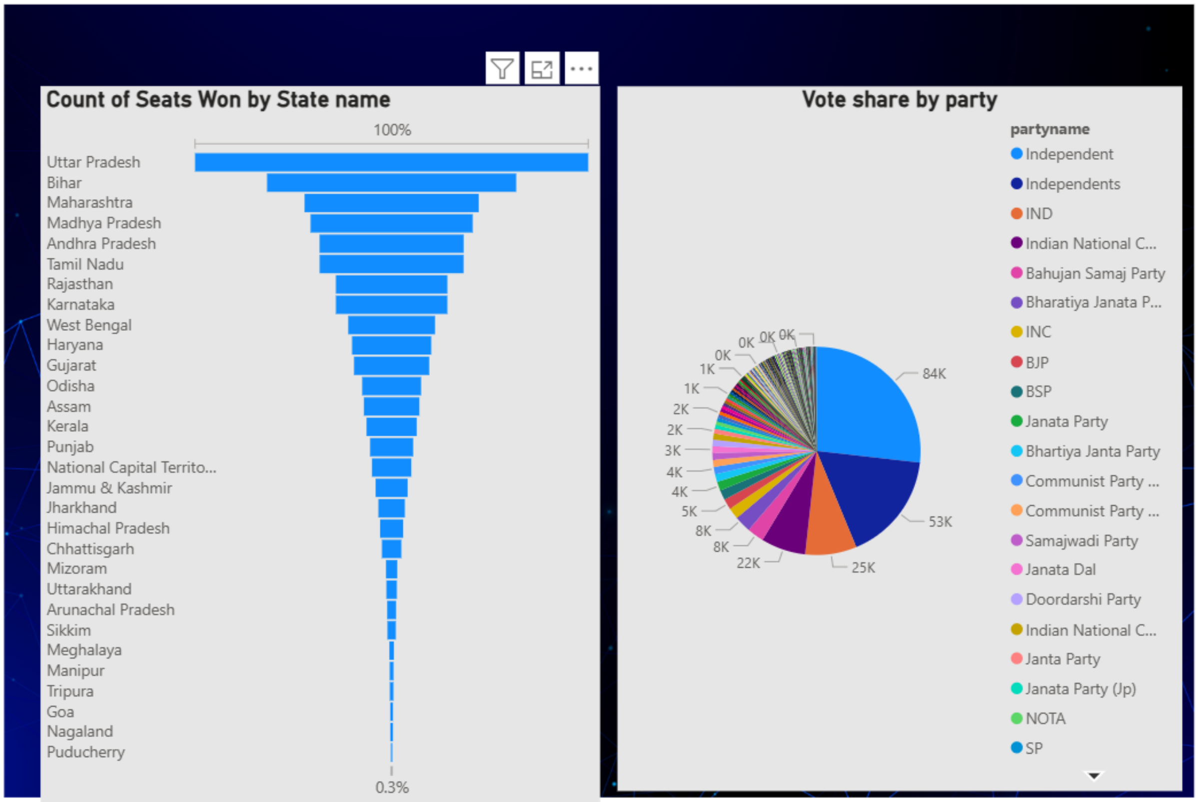
7. Outcome of Project Execution The successful completion of the ElectViz project resulted in a comprehensive, 5-page interactive Power BI dashboard. The tool effectively consolidates and visualizes decades of complex election data, providing a powerful resource for media, analysts, and the public to derive clear and accurate insights.

6. Snapshots / Screenshots




Total Votes by year
200M
150M
100M
50M
0M
Total Votes
1980
1990
2000
2010
year
Voter Turnout % by State Name
25.87971610532531 (8.69%)
19.971934911143034 (6.71%)
18.66597578494441 (6.27%)
17.0325370007249... (5.72%)
16.95791739766642 (5.7%)
15.546996501060361 (5.22%)
14.40047597463561 (4.84%)
13.809615679523521 (4.64%)
13.31728653626921 (4.47%)
11.625477048357553 (3.9%)
8.640597246923875 (2.9%)
8.22539869947225 (2.76%)
7.211707960402255 (2.42%)
7.002880938265984 (2.35%)
6.8833690386296 (2.31%)
6.448892099316375 (2.17%)
6.228472081579664 (2.09%)
6.0065019199508 (2.02%)
5.735651979400156 (1.93%)
5.376316241662735 (1.81%)
State Name
Nagaland
Tripura
Arunachal Pradesh
Manipur
Mizoram
Meghalaya
Goa
Puducherry
Sikkim


Top Party
Independent
Top State
Uttar Pradesh
Count of Seats Won by year
Increase
Decrease
Total
150K
100K
50K
0K
Count of Seats Won
1985 1990 1993 1995 1996 1991 1980 1977 2014 1989 2008 2009 2012 2013 2007 2002 2005 2004 2011 1998 2006 1994 2003 2000 1978 1999 2001 1987 1983 2010 1982 1984 2015 1997 1992 1979 Total
year



Sate-wise Party Seat Distribution

State Name	A S P	Aa S P	AAAP	AACP	Aadivasi Sena Party	AAMP	AAP	AASAP	AaSP	ABAS	ABCD(A)	ABDBM	ABGL	ABGP	ABHM	AB
Andhra Pradesh			79													
Arunachal Pradesh																
Assam																
Bihar																
Chhattisgarh		1												2	1	
Goa																
Gujarat					2											
Haryana																
Himachal Pradesh																
Jammu & Kashmir																
Jharkhand						1									2	
Karnataka															1	
Kerala																
Madhya Pradesh									8					16	3	
Maharashtra															11	
Manipur																
Meghalaya																
Mizoram																
Nagaland																
National Capital Territory Of Delhi		2	59	4				1							5	
Odisha			57													
Puducherry																
Punjab																
Rajasthan							2				4				11	
Sikkim																
Tamil Nadu															6	
Tripura																
Uttar Pradesh	4	1								1		1			6	
Total	**4**	**4**	**195**	**4**	**2**	**1**	**2**	**1**	**8**	**1**	**4**	**1**	**2**	**18**	**47**	

7. Challenges Faced

During the course of this internship, the team encountered several technical, operational, and communication challenges that required problem-solving and collaboration to overcome.

1. Technical Challenges

Data Cleaning and Consistency: The most significant technical hurdle was the raw dataset itself. The partyname column contained over 1,637 unique entries, including duplicates, abbreviations, and different spellings for the same party (e.g., "Independent" vs. "IND").

Resolution: We used Power Query Editor extensively to clean and transform this data. We standardized party names by grouping similar entries and merging duplicates. This was a time-consuming but critical step to ensure the accuracy of visuals like "Seats Won by party Name" and the "Total Parties" (1637) KPI.

Data Volume and Dashboard Performance: The dataset was massive, covering elections from 1977 to 2014 and representing over 3044 million votes. A key challenge was ensuring the 5-page dashboard remained fast and responsive when users applied filters.

Resolution: We optimized the data model by creating efficient relationships and avoiding unnecessary columns. We relied on DAX measures for calculations rather than creating large calculated columns, which improved the dashboard's load time and interactivity.

DAX Measure Complexity: Creating accurate KPIs was challenging. For example, the "Voter Turnout %" (5.52%) was not a simple average; it had to be calculated as the sum of all votes divided by the sum of all eligible electors.

Resolution: The team referred to DAX documentation and tutorials to write correct and robust measures. We used functions like SUMX and DIVIDE to ensure calculations were accurate and handled potential "divide by zero" errors.

2. Operational Challenges

Coordinating Work Across Multiple Dashboards: With four team members working on five different dashboard pages, ensuring a unified final product was difficult.

Resolution: We assigned clear ownership: one member focused on data cleaning and modeling, while others took responsibility for specific pages (e.g., Main Dashboard, Historical Trends, Matrix page). We used a shared workspace to maintain version control of the .pbix file.

Balancing Internship with Academic Workload: Managing the weekly project deliverables while keeping up with regular academic classes and assessments was a consistent challenge.

Resolution: The team created a detailed weekly plan based on the "Timeline Overview." We divided tasks evenly and held short daily check-in meetings to track progress and support any member who was falling behind.

3. Communication Challenges

Ensuring Visual Consistency: A major communication challenge was making sure all five dashboards looked and felt like a single, cohesive report (e.g., same font sizes, color themes, and slicer placement).

Resolution: We created a template page with the standard dark theme, header, and slicer layout. Team members duplicated this template to build their assigned pages, which ensured the final product was visually aligned and professional.

8. Learnings & Skills Acquired

- **Technical Skills:**
  - Proficient use of **Power BI** for creating complex, multi-page dashboards.
  - Advanced **Power Query** skills for data cleaning and transformation.
  - Practical application of **DAX** to create meaningful KPIs from raw data.

- **Analytical Skills:**
    - Gained domain knowledge in election data analysis.
    - Learned to identify political trends, party performance metrics, and voter behavior patterns.
    - Enhanced ability to translate complex data into a simple visual story.
- **Soft Skills:**
    - **Teamwork:** Collaborated effectively to design, build, and test the dashboard.
    - **Problem-Solving:** Worked together to overcome data cleaning and modeling challenges.
    - **Communication:** Developed a report that clearly communicates insights to a target audience (media).

9. Testimonials from team

P. Pradeep Reddy: "Working on the ElectViz project was an incredible and enriching experience. Applying Power BI to such a large and complex dataset (3044M votes) was challenging but very rewarding. I learned so much about the practical side of data cleaning and performance optimization. I'm proud of the interactive tool we built, which can genuinely help media reporters find data-driven stories. I am deeply grateful to our mentor, Mrs. Nithyasri S J for her continuous guidance."

Pavithiraa S: "This project was a great practical learning opportunity. I particularly enjoyed the challenge of designing the 'State-wise Party Seat Distribution' matrix (Dashboard 5). It required careful data modeling to make it work correctly with all the slicers. This internship deepened my understanding of how crucial a clean and logical data model is for a successful dashboard. I am sincerely thankful to my mentor for their support."

Vitta Hem Kumar: "The ElectViz project allowed me to see the real-world power of data analytics. The biggest challenge for me was tackling the 1,637 unique party names in Power Query. Transforming that messy data into a clean, usable dataset was a major accomplishment for our team. I improved my data cleaning and transformation skills significantly. My thanks to the entire team and our mentor for this experience."

Charan Teja: "I am thrilled with what our team accomplished. I focused on building the main dashboard (Dashboard 1) and creating the core KPIs using DAX. Seeing the 'Total Votes by year' trend line come to life was fascinating. This internship project taught me the importance of teamwork, clear communication, and designing a dashboard with a specific audience in mind. I'm grateful for the mentorship we received."

10. Conclusion

This internship has been an enriching and transformative journey that strengthened both our technical and professional capabilities. Throughout this program, we gained deep exposure to the world of data analytics, visualization, and business intelligence, while also learning to think critically and collaboratively to solve real-world problems.

The project, "ElectViz: Indian State Election Analysis for Media," provided an opportunity to tackle a large-scale, complex dataset. By analyzing historical election results, party performance, and voting trends, our team was able to derive meaningful insights that reflect the intricate nature of India's political landscape.

This experience taught us how to convert raw, and often messy, data (with over 1,600 party names) into actionable intelligence. The development of a 5-page interactive Power BI dashboard demonstrated how visualization can simplify complex datasets, helping media professionals and analysts identify trends, compare state-wise performance, and make informed, data-driven observations.

Beyond technical learning, this internship emphasized the importance of team coordination, time management, and communication. Working in a virtual environment required us to collaborate effectively, share knowledge, and take ownership of individual responsibilities. Each phase—from data cleaning in Power Query to building the final matrix—involved structured planning, problem-solving, and continuous improvement.

Overall, this internship was not just a project but a journey of learning, innovation, and collaboration. It has inspired us to continue exploring data analytics as a powerful tool for driving clarity and transformation in any industry.

11. Acknowledgements

We would like to express our heartfelt gratitude to our internship program, Infosys Springboard, for providing this invaluable opportunity to gain practical exposure in the field of Data Analytics and Business Intelligence. The structured learning modules, hands-on projects, and professional mentorship created a truly immersive learning environment.

A special note of appreciation is extended to our mentor, Mrs. Nithyasri S J for their constant guidance, encouragement, and insightful feedback throughout the internship. Their mentorship played a crucial role in helping us refine our dashboard design, ensure data accuracy (especially with the complex dataset), and present our findings in a professional and impactful manner.

We are also thankful to the program coordinators and the learning community for providing high-quality resources, tutorials, and continuous support. These materials greatly enhanced our understanding of Power BI, DAX, and advanced data visualization techniques, allowing us to build a meaningful and well-structured analytical solution.

Finally, we express our sincere gratitude to everyone involved in fostering this culture of innovation and learning. The experience gained through this internship will remain a cornerstone of our academic and professional journey.